Marketing A/B Test: Ad Performance Analysis

May 30, 2026

1. Introduction

This project focuses on A/B testing on marketing data. A/B testing is a method used to compare two versions of a variable to determine whether there is a statistically significant difference in their performance. You can find the code in this repository, and the mathematical formulas used in the analysis are provided in the Appendix for those who are interested in the underlying methodology.

I completed this project to reinforce my understanding of statistical testing, which I previously learned in university. If you notice any issues or errors in my calculations, please feel free to reach out, as I am still in the process of learning.

2. Data Acquisition

Our dataset is sourced from Kaggle and is named *marketing_AB.csv*, containing information related to a marketing advertisement campaign. The dataset has been cleaned (refer to the Jupyter Notebook for details) and consists of 588,101 rows and 6 columns. The data includes the following variables:

- User_id: Unique identifier for each user.
- Test_group: Indicates whether the user was assigned to the ad or PSA group.
- Converted: Indicates whether the user converted (1) or did not convert (0).
- Total_ads: Total number of advertisements viewed by the user.
- Most_ads_day: Day of the week on which the user viewed the most advertisements.
- Most_ads_hour: Hour of the day during which the user viewed the most advertisements.

	user_id	test_group	converted	total_ads	most_ads_day	most_ads_hour
0	1069124	ad	0	130	Monday	20
1	1119715	ad	0	93	Tuesday	22
2	1144181	ad	0	21	Tuesday	18
3	1435133	ad	0	355	Tuesday	10
4	1015700	ad	0	276	Friday	14
...	...	...	...	...	...	...
588096	1278437	ad	0	1	Tuesday	23
588097	1327975	ad	0	1	Tuesday	23
588098	1038442	ad	0	3	Tuesday	23
588099	1496395	ad	0	1	Tuesday	23
588100	1237779	ad	0	1	Tuesday	23

Table 1: Sample of Dataset

3 Exploratory Data Analysis (EDA)

EDA was conducted to better understand the dataset, identify key patterns, and examine relationships between variables.

3.1 Test Group Distribution and Conversion Rates

In Fig 1, the ad group is substantially larger than the psa group. Despite this imbalance, the conversion rate for the ad group is noticeably higher than that of the psa group, suggesting that exposure to actual ads has a meaningful lift on conversions.

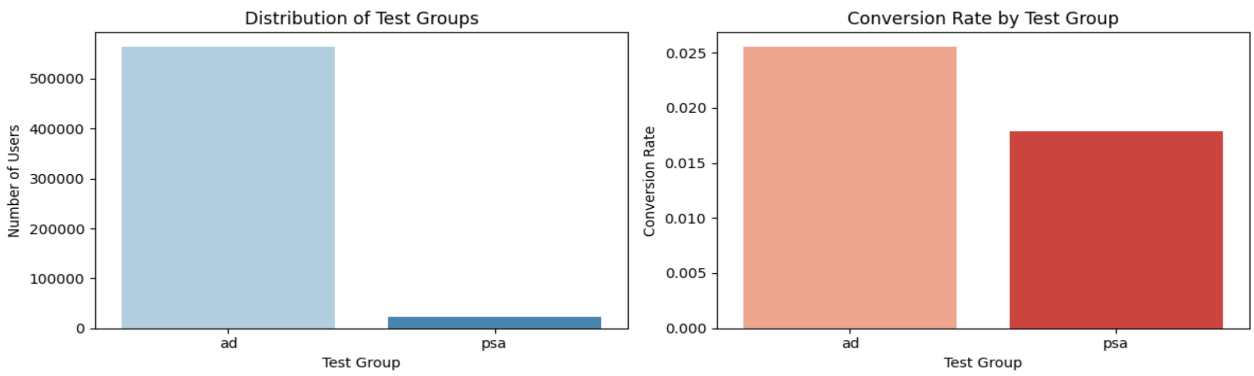

Fig 1: User Distribution and Conversion Rate by Test Group

3.2 Ad Exposure and Conversions across Days of the Week

In Fig 2, the number of users exposed to the most ads is fairly consistent across all seven days, hovering between 77,000 and 92,000. In terms of conversions, Monday stands out with the highest conversions despite not having the most users, while Friday has the largest audience but one of the lower conversion counts.

A possible reason is that people who see ads on Monday are likely in a "fresh week, let's get things done" mind-set, making them more action-oriented and decisive. Alternatively, advertisers may allocate higher budgets or better-targeted ads on Mondays to capitalise on this. Another contributing factor could be that many users research products over the weekend on their phones, then convert on Monday when they are back at their desks with easier access to payment methods (desktop vs mobile)

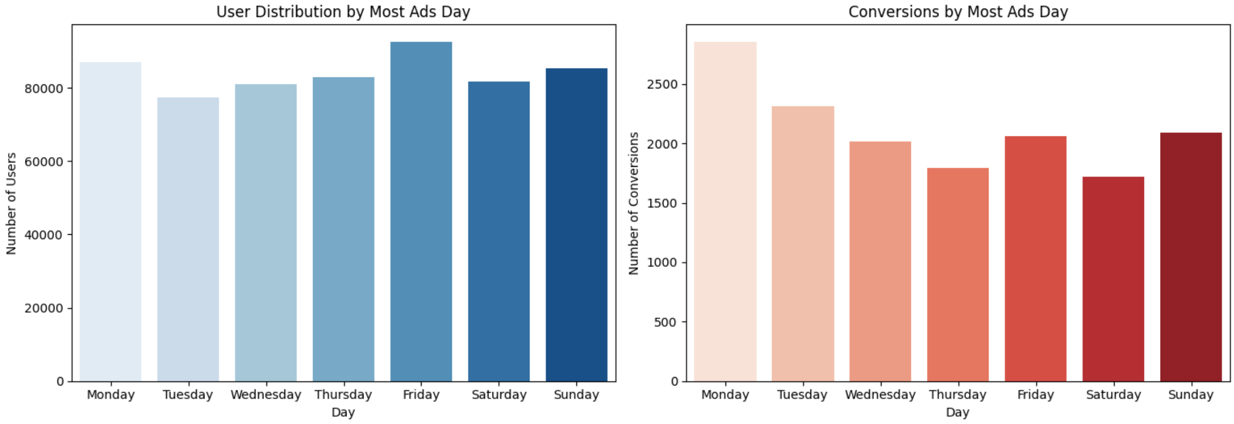

Fig 2: User Distribution and Number of Conversions by Most Ads Day

3.3 Ad Exposure and Conversions across Hours of the Day

In Figure 3, both charts follow a very similar pattern. Ad activity picks up from around 9am, peaks in the early afternoon between 12pm and 3pm, and gradually tapers off through the evening. Conversions closely mirror this trend, suggesting that the conversion rate is fairly stable across the day and is largely driven by the volume of users reached at each hour.

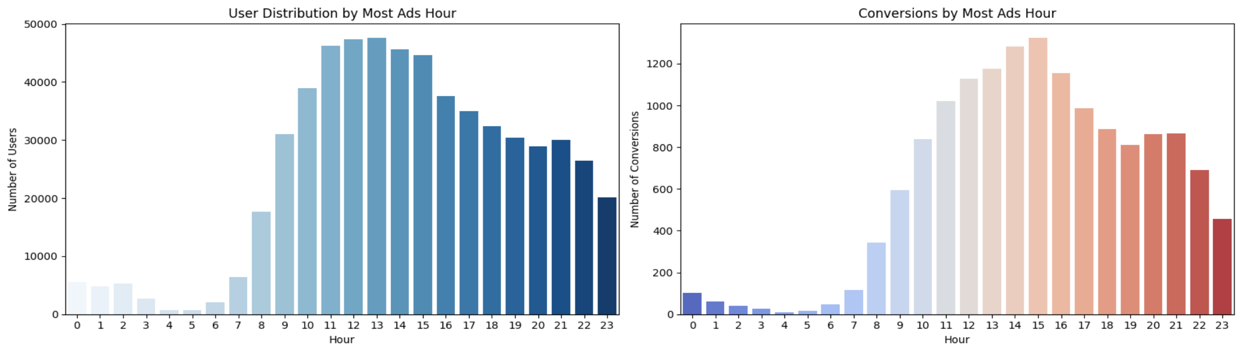

Fig 3: User Distribution and Number of Conversions by Most Ads Hour

3.4 Distribution of Total Ads Seen per User

In Figure 4, we remove the outliers to have a better visualization of the distribution seen in Fig 5. Both the ad and psa groups show heavily right-skewed distributions, with most users seeing a relatively small number of ads while a minority are exposed to a much larger count.

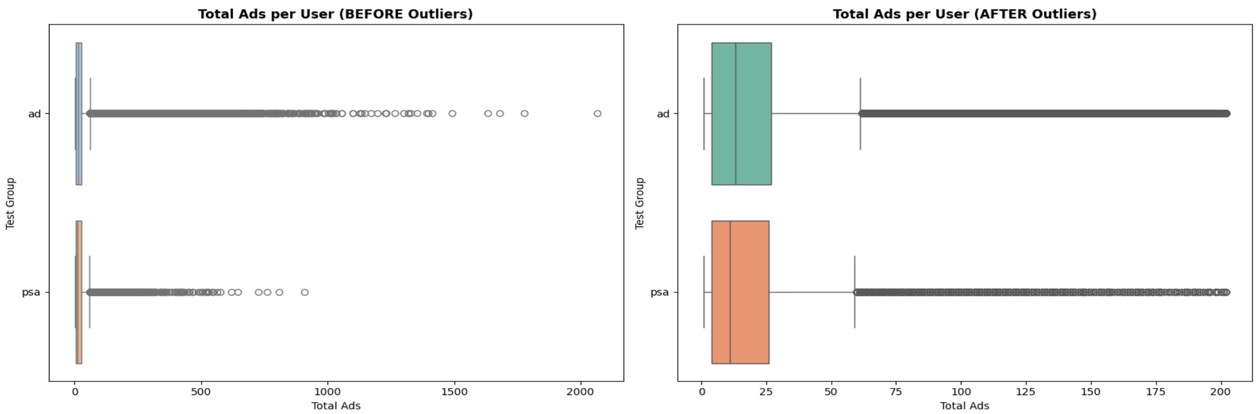

Figure 4: Total Ads per User Before and After Outlier Removal by Test Group

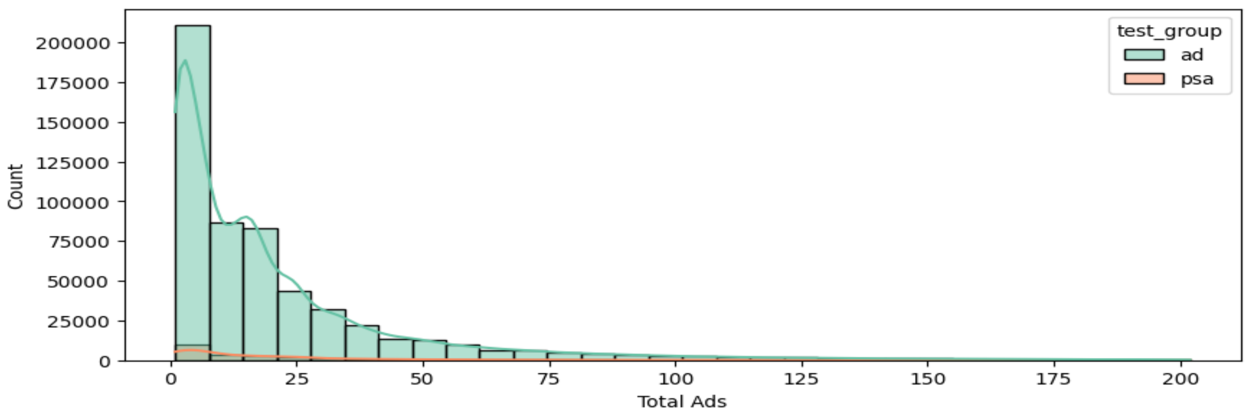

Figure 5: Distribution of Total Ads Seen per User by Test Group

4. Statistical Analysis

To ensure our A/B testing results are not merely coincidental or based on visual inspection alone, we use statistical tests to determine whether the observed differences are statistically significant and commercially actionable.

4.1 To evaluate whether the advertisement truly increases the conversion rate

Null Hypothesis (H_0): The conversion rate in the ad group is less than or equal to that of the PSA group($p_{ad} \leq p_{psa}$)

Alternative Hypothesis (H_1): The conversion rate in the ad group is higher than that of the PSA group ($p_{ad} > p_{psa}$)

A one-tailed two-sample z-test for proportions is used, with the following observed data:

Test Group	Conversions	Total Users
ad	14423	564577
psa	420	23524

Terminology

- p-value: The probability of observing results at least as extreme as ours, assuming H_0 is true
- ci_lower_pp / ci_upper_pp: The 95% confidence interval of the lift in percentage points
- lift_pp: The difference in conversion rates between the ad and psa groups, in percentage points

Results

z_stat	p_value	ci_lower_pp	ci_upper_pp	lift_pp
7.370078	8.526404e-14	0.595093	0.943397	0.769245

Bootstrap Results: Bootstrap resampling repeatedly resamples users with replacement from both groups, recomputing the lift and p-value each time.

p_value	ci_lower_pp	ci_upper_pp	lift_pp
0.000	0.603687	0.946307	0.769245

Conclusions

Since the p-value is far below 0.05, we reject H_0. There is statistically significant evidence that advertisement exposure increases the conversion rate, with an estimated lift of 0.769 percentage points and a 95% confidence interval of (0.595, 0.943). This is further confirmed by bootstrap resampling, which yields a consistent lift estimate with a confidence interval that does not include zero, indicating the result is robust and not driven by sampling variability.

4.2 To Evaluate Whether the Advertisement Effect Varies Across Days

To assess whether users were randomly and evenly distributed between the ad and PSA groups across different days, and whether the conversion effect varies by day, a Chi-square test of independence is conducted.

Null Hypothesis (H_0): The test group and most_ads_day are independent (there is no association between group assignment and day distribution).

Alternative Hypothesis (H_1): The test group and most_ads_day are not independent (there is an association between group assignment and day distribution).

most_ads_day	ad	psa
Friday	88805	3803
Monday	83571	3502
Saturday	78802	2858
Sunday	82332	3059
Thursday	79077	3905
Tuesday	74572	2907
Wednesday	77418	3490

Chi-Square Test Result: p-value = 4.85e-48

Since the Chi-square test only tells us that an association exists but not where, a post-hoc analysis using two-sample proportion z-tests with FDR correction is conducted for each day individually.

Terminology

- p-value (FDR): Running multiple z-tests increases the probability of false positives. FDR (False Discovery Rate) correction adjusts the p-values to control for this— refer to the Appendix for the full formula.
- Lift (%): The percentage increase in conversion rate of the ad group relative to the psa group for that specific day.
- Significant: Whether the adjusted p-value falls below 0.05, indicating a statistically meaningful difference for that day

day	p_value	p_value_fdr	ci_lower	ci_upper	lift_percent	significant
Monday	5.078178e-04	0.001185	0.005088	0.015272	47.355277	True
Tuesday	6.634401e-07	0.000005	0.010826	0.019939	110.690914	True
Wednesday	3.764133e-04	0.001185	0.004777	0.013400	60.894460	True

day	p_value	p_value_fdr	ci_lower	ci_upper	lift_percent	significant
Thursday	5.547506e-01	0.554751	-0.003603	0.005507	6.953197	False
Friday	1.156888e-02	0.016196	0.001522	0.009863	37.797113	True
Saturday	7.483818e-03	0.013097	0.002209	0.011153	52.235413	True
Sunday	1.571861e-01	0.183384	-0.001736	0.008618	19.542962	False

Conclusions

- Based on the Chi-square test of independence, the null hypothesis is rejected, indicating that there is an association between test group (ad vs PSA) and most_ads_day. This suggests that the distribution of users across days is not fully independent of group assignment.
- Post-hoc analysis using two-sample proportion z-tests with FDR correction shows that the advertisement effect varies across days of the week. Significant positive conversion lifts are observed on Monday, Tuesday, Wednesday, Friday, and Saturday, while no statistically significant differences are found on Thursday and Sunday.
- The confidence intervals for significant days are generally above zero, indicating a positive and measurable effect of the advertisement during those periods.

4.3 To further evaluate the effect of advertisement on conversion and whether this effect differs across days using a two-way ANOVA model.

	df	sum_sq	mean_sq	F	P-Value(PR>F)
C(test_group)	1.0	1.336325	1.336325	54.361228	1.670394e-13
C(most_ads_day)	6.0	10.068857	1.678143	68.266244	2.589796e-85
C(test_group):C(most_ads_day)	6.0	0.430494	0.071749	2.918723	7.574369e-03
Residual	588087.0	14456.543895	0.024582	NaN	NaN

Terminology

- df (Degrees of Freedom): The number of independent values that went into estimating each effect.
- sum_sq (Sum of Squares): The total variation in conversion rates explained by each factor.
- mean_sq (Mean Square): Sum of squares divided by degrees of freedom.
- F-statistic: The ratio of mean_sq of the factor to mean_sq of the residual.
- Residual: The variation in conversion rates not explained by any factor in the model

Conclusions

The ANOVA results further confirm these findings from 4.1 and 4.2. There are significant main effects of both test group and day of week on conversion rates, indicating that the advertisement influences conversion overall and that conversion behaviour differs across days. Importantly, the significant interaction effect between test group and day of week suggests that the effectiveness of the advertisement varies depending on the day.

5. Business Recommendations

Based on the findings in Section 4, the following recommendations are proposed to maximise the effectiveness of the advertisement campaign.

- **Continue investing in ads.** The ad group consistently outperforms the PSA group across all statistical tests. The evidence supports continued and increased investment in advertisement exposure.
- **Reallocate budget by day.** Tuesday, Wednesday, and Saturday show the strongest conversion lifts, while Thursday and Sunday show no statistically significant effect. Shifting budget away from Thursday and Sunday toward these high-performing days would reduce wasted spend and improve overall conversion efficiency.
- **Concentrate ad delivery between 12pm and 3pm.** Ad activity and conversions both peak during this window, with activity picking up from 9am and tapering off through the evening. Combining time-of-day targeting with the high-lift days above would maximise reach during periods of highest purchase intent.
- **Implement ad frequency capping.** Based on EDA findings, a segment of users were exposed to an excessive number of ads. Capping ad frequency would prevent ad fatigue and ensure budget is directed toward reaching new users rather than repeatedly targeting the same ones.

6. Conclusion

This project applied A/B testing and statistical analysis to evaluate the effectiveness of an advertisement campaign. The results show that advertisement exposure significantly increases conversion rates, with the effect varying across days of the week and peaking mid-week.

The business recommendations derived from these findings are targeted and actionable, focusing ad spend on high-performing days, concentrating delivery during peak hours, and managing ad frequency to avoid diminishing returns. Together, these steps could meaningfully improve the efficiency and return on investment of future campaigns.

7. Appendix

FDR Correction (Benjamini-Hochberg Procedure)

This was my first time coming across this, so I used an example to help understand it better.

Say I flip a fair coin 7 times. Each flip has a 5% chance of landing heads by pure luck. The more flips I do, the higher the chance that at least one comes up heads just by chance — not because the coin is biased. Running 7 hypothesis tests (as in Section 4.2) follows the same idea. Each test has a 5% chance of a false positive, so the more tests run, the more likely at least one false positive slips through by luck.

The procedure for adjusting the raw p-values is as follows:

Step 1: Rank all p-values from smallest to largest:

$$p_1 \leq p(2) \leq \cdots \leq p_m$$

Step 2: For each ranked p-value, compute the adjusted p-value:

$$p_{adj(i)} = p_i \times \frac{m}{i}$$

- m = total number of tests
- i = rank of the p-value(1=smallest, m = largest)

Step 3: Enforce monotonicity from the largest rank downward:

$$p_{adj_i} = \min(p_i \times \frac{m}{i}, p_{ads(i+1)})$$

Step 4: Reject H_0 for all tests where:

$$p_{adj_i} \leq \propto = 0.05$$

Referring to Section 4.2, m=7m = 7 m=7 (one test per day), so the most significant result is barely penalised while the least significant receives the full 7-fold adjustment.

When should FDR correction be used?

FDR correction is needed whenever you run multiple hypothesis tests on the same dataset at once — like we did in Section 4.2, where we ran one z-test per day. Without it, you risk declaring a result significant when it's just a fluke. If you are only running a single test like in Section 4.1, no correction is needed.

Derivation of two-proportion z-test.

$X \sim \text{Bernoulli}(p)$ where $0 \leq p \leq 1$

$Pr(X = x) = p^{x}(1-p)^{1-x}$, for $x = 0, 1$

$E[x] = \sum x \, Pr(X = x)$

$= 0 \, Pr(x = 0) + Pr(x = 1)$

$= p^{1}(1-p)^{1-0} = p$

$Var(x) = E[x^2] - (E[x])^2$

$= \sum x^2 Pr(x = x) - p^2$

$= Pr(x = 1) - p^2$

$= p - p^2$

$= p(1-p)$

for a sample of size n, $\{x_1, \ldots, x_n\}$, p is estimated by the sample proportion

$$\hat{p} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

By central limit theorem, the sample mean of any iid random variables converges to $N(0,1)$ as $n \to \infty$

$E[\hat{p}] = E\left[\frac{1}{n}\sum_{i=1}^{n} x_i\right]$

$= \frac{1}{n}\sum_{i=1}^{n} E[x_i]$

$= \frac{1}{n} \cdot np$

$= p$

$Var(\hat{p}) = Var\left(\frac{1}{n}\sum_{i=1}^{n} x_i\right)$

$= \frac{1}{n^2} Var\left(\sum_{i=1}^{n} x_i\right)$

$= \frac{1}{n^2}\sum_{i=1}^{n} Var(x_i)$

$= \frac{1}{n^2} \cdot n(p)(1-p)$

$= \frac{p(1-p)}{n}$

So $\hat{p} \sim N\left(p, \frac{p(1-p)}{n}\right)$ as $n \to \infty$

If two proportion groups are independent, find distribution of $\hat{p}_1 - \hat{p}_2$

$E[\hat{p}_1 - \hat{p}_2] = E[\hat{p}_1] - E[\hat{p}_2]$

$= p_1 - p_2$

$Var(\hat{p}_1 - \hat{p}_2) = Var(\hat{p}_1) + Var(\hat{p}_2)$

$= \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}$

hence $\hat{p}_1 - \hat{p}_2 \sim N\left(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right)$

$$\Leftrightarrow Z = \frac{(\hat{p}_1 - \hat{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

$\Leftrightarrow$ under H_0, we assume $p_1 = p_2$, so $(p_1 - p_2) = 0$

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Derivation of Chi-Square Test

We have a contingency table with r rows and C columns. The observed count in cell(i,j) is O_{ij} and the row and column totals are R_i and C_j respectively with grand total N

	$Y=1$	$Y=2$	$\dots$	$Y=C$	Row Total
$X=1$	O_{11}	O_{12}	$\dots$	O_{1c}	R_1
$X=2$	O_{21}	$\ddots$		$\vdots$	R_2
$\vdots$	$\vdots$		$\ddots$	$\vdots$	$\vdots$
$X=r$	O_{r1}	$\dots$	$\dots$	O_{rc}	R_r
Col Total	C_1	C_2	$\dots$	C_c	N

Expected values for each cell in the table

$$\Rightarrow \frac{(\text{Row Total } R_i) \times (\text{Col Total } C_j)}{\text{Grand Total } (N)}$$

$$= \frac{R_i C_j}{N}$$

For each cell, we measure how far the observed count is from expected count:

$$O_{ij} - E_{ij}$$

Then we square it to remove potential negative sign and normalise by E_{ij} to account for scale:

$$\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Next we sum over all cells and this gives us the chi-square statistics

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

Reason: If I have $R_1, \dots, R_r$ once I know $c-1$ cells, the last cell automatically determined because all entries must sum R_i

For Degrees of Freedom, $df = (r-1)(c-1)$ Logic applies to $C_1, \dots, C_c$ too.

P-value for chi square test is $Pr(\chi_v^2 > \chi^2)$

Derivation of 1 way ANOVA Test.

Suppose we have K populations with

- means $\mu_1, \mu_2, \dots \mu_k$ and
- Variances $\sigma_1^2, \sigma_2^2 \dots, \sigma_k^2$ respectively

Samples denoted as follows

$\{x_{11}, x_{12}, x_{13}, \dots x_{1n_1}\}$ from population 1

$\vdots$

$\{x_{k1}, x_{k2}, x_{k3}, \dots x_{kn_k}\}$ from population k

All Samples x_{ij} are independent of each other and $\sigma_1^2 = \sigma_2^2 = \dots \sigma_k^2 = \sigma^2$

for each group i,

$$\bar{x}_i = \frac{1}{n_i}\sum_{j=1}^{n_i} x_{ij} \quad \text{and} \quad S_i^2 = \frac{1}{n_i-1}\sum_{j=1}^{n_i}(x_{ij}-\bar{x}_i)^2$$

Within-group Mean Square (MS): Measures how much observations vary within each group

$$S_w^2 = \frac{1}{n-k}\sum_{i=1}^{k}\sum_{j=1}^{n_i}(x_{ij}-\bar{x}_i)^2 = \frac{1}{n-k}\sum_{i=1}^{k}(n_i-1)S_i^2$$

Between Group Mean Square (MS): Measures how much the group mean differ from the overall mean

$$S_b^2 = \frac{1}{k-1}\sum_{i=1}^{k}\sum_{j=1}^{n_i}(\bar{x}_i-\bar{x})^2 = \frac{1}{k-1}\sum_{i=1}^{k} n_i(\bar{x}_i-\bar{x})^2$$

If $H_0: \mu_1 = \mu_2 = \dots = \mu_k$ is true, then we should expect S_b^2 to be Small.

$$E[S_b^2] = \; \vdots \; \sigma^2 + \underbrace{\frac{\sum_{i=1}^{k} n_i(\mu_i-\mu)^2}{k-1}}_{\text{group effect}}$$

skip steps cause its too long

$E[S_w^2] = \sigma^2$

under H_0: group effect $= 0$, so $E[S_b^2] = E[S_w^2] = \sigma^2$

under H_1: $E[S_b^2] > E[S_w^2]$

Hence the F-Statistic: $F = \frac{S_b^2}{S_w^2} \sim F(k-1, N-k)$

under $H_0: F \approx 1$

Under H_1 $F > 1$